Project Name	Project 2 - NIM - Automated Thematic Analysis
Online team meeting	https://fau.zoom-x.de/j/64229233763?pwd=MboXckMTe5aVIBRIwvO71OgqjXJ7zb.1
Production system (if any)	...
Test system (if any)	...
GitHub repository	https://github.com/amosproj/amos2026ss02-automated-thematic-analysis
GitHub feature board	https://github.com/orgs/amosproj/projects/98
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/102
Position of T-shirt logo	Upper Torso Right Small
Link to logo for black T-shirt	https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg
Link to logo for white T-shirt	https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg
Additional materials	...
Team mailing list	oss-amos-proj2@lists.fau.de **(Emails sent here get quarantined by the TU Berlin Spam Filter)**

Last Name	First Name	GitHub User Name	Email Address
Kahvecioglu	Selin	slnkahveci	kahvecioglu@campus.tu-berlin.de
de Sa Filho	Fabiano	fdesafilho	fabiano.de.sa@fau.de
Ammann	Paul	Wecoreator	p.ammann@campus.tu-berlin.de
Schmidt	Alexander	alexanderschmi	a.schmidt.2@tu-berlin.de
Wendlandt	Julian	julianlucawendlandt-jpg	julian.luca.wendlandt@campus.tu-berlin.de
Müermann	Ole	ClimbTheGrind	ole.mueermann@fau.de
Klarer	Paul	paulKlarer	paul.linus.klarer@fau.de
Mantsch	Christoph	Christoph-Mantsch	christoph.cm.mantsch@fau.de
Annus	Mohammad	annus3	mohammad.annus@fau.de
Rasul	Hamid	hrasul	hamid.rasul@fau.de

		Product Owner					
#	Meeting Day	Review	Planning	Software Developer	Release Manager	Scrum Master	Comment
1	2026-04-15	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	N/A	Christoph Mantsch	
2	2026-04-22	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Paul Klarer	Christoph Mantsch	
3	2026-04-29	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Paul Klarer	Christoph Mantsch	
4	2026-05-06	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Selin Kahvecioglu	Christoph Mantsch	
5	2026-05-13	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Ole Müermann	Christoph Mantsch	
6	2026-05-20	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Ole Müermann	Christoph Mantsch	
7	2026-05-27	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Hamid Rasul	Christoph Mantsch	Mid-term due
8	2026-06-03	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Selin Kahvecioglu	Christoph Mantsch	
9	2026-06-10	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Hamid Rasul	Christoph Mantsch	
10	2026-06-17	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Paul Ammann	Christoph Mantsch	
11	2026-06-24	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Mohammad Annus	Christoph Mantsch	
12	2026-07-01	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Mohammad Annus	Christoph Mantsch	
13	2026-07-08	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Paul Ammann	Christoph Mantsch	
14	2026-07-15	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Julian Wendlandt	Christoph Mantsch	Demo day!
15	2026-07-22	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Julian Wendlandt	Christoph Mantsch	Retrospective
Product owners, software developers, and Scurm Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

Goals	Having Fun
	Satisfy NIM
	Program an automated thematic analysis of interview transcripts
Meeting norms	Joining the meeting is mandatory
	Always be on time (up to 10 min late is fine if short message on discord)
	Follow the timeline
	Take notes in each meeting
	Happiness Index after Sprint retrospective
	Each meeting takes 1:30 hours
Working norms	Tell us if/when you are working on a task
	We encourage pair programming
Coordination norms	Self assign issues you are opening
	Do not force push
	Use PEP 8 as style (configure your IDEs)
	Follow https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow
	Once every workday check discord
	If someone is blocked share with the team
Communication norms	Criticism should be productive
	Language should be respectful at all time (written and oral)
Consideration norms	Discuss disagreement openly
	Follow the NDA and data privacy
	Try not to induce bias in our project
Cont. improvement norms	Finish work until tuesday evening 20:00 PM
	At the end of the meeting discuss happyness score / what could be done better / what was good
Rewards	Have a happy meeting
Sanctions	Write a poem saying you are sorry
	After multiple or serious offenses escalate to prof
Signatures	
Scrum Master	Christoph Mantsch
Product owner	Alexander Schmidt
Product owner	Fabiano de Sa Filho
Software developer	Ole Müermann
Software developer	Julian Wendlandt
Software developer	Hamid Rasul
Software developer	Paul Klarer

Software developer	Paul Ammann
Software developer	Selin Kahvecioglu
Software developer	Mohammad Annus

Product Vision	Project Mission
LLMs are a fantastic tool to interview research subjects at scale, which previously had to be done by human researchers. This method generates unstructured interview dialogues which must then be processed and analysed manually. The goal of this project is to use LLMs to also automate the analysis part, enabling policy researchers to feed their interview corpus to our tool and obtain not only the thematic analysis, but also relevant insights derived from the demographic data of the respondents. This way, we seek to bridge the gap between quantitative and qualitative policy research.	To engineer a secure, open-source tool that ingests thousands of subject interviews about a policy topic and produces an accurate thematic analysis breakdown. The core is a multi-stage LLM pipeline that extracts quotes, codes them into themes, clusters and ranks those themes, and cross-references them against respondent demographic dimensions. Results are surfaced through a lightweight frontend and exportable in structured formats. We also provide direct reference back to the corpus, so researchers can see which interviews most strongly manifest a particular theme or demographic combination. We build on QuaLLM by incorporating this demographic layer, inferring how discovered themes distribute across the profile of research subjects, and surfacing patterns that a purely manual process would likely miss.

Term	Definition
Corpus / Dataset	The 1,058 LLM-led interviews on the economic impact of AI provided by the Industry Partner. Format: JSON
Demographic Data	Categorical variables describing each subject in the Corpus; has been partially sanitized by Industry partner, but must be handled with care. Absolutely should not be downloaded or uploaded to third parties. Format: CSV
Thematic Analysis	The qualitative method of identifying and analyzing patterns of meaning (themes) in a dataset. These themes should be followed by a short description and some quantitative measures of prevalence in the dataset.
QuaLLM	The specific LLM-based framework serving as the baseline architecture for our project. Suggested by Dr. Kaiser, can be seen on Rao et al.

Sprint #	Sprint goal
1	Meeting with industry partner, designing the logo, setting up the project
2	Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully passes text through a basic LLM endpoint.
3	Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and llm communication
4	Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Readme and writting a proper Build script.
5	Closing up all features from previous sprints, finalizing the frontend so that we have a working demo for Industry Partner.
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	

Sprint #	Story Points Realized
1	N/A
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			0	0		
Sprints						
1			0	0	0	0
2			0	0	0	0
3			0	0	0	0
...				0		0
Features						
1						
2						
3						
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			0	0		
Sprints						
1			0	0	0	0
2			0	0	0	0
3			0	0	0	0
...				0		0
Features						
1						
2						
3						
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

#	Feature Definition of Done	Sprint Release Definition of Done	Project Release Definition of Done

Type	Link / reference

#	Context	Name	Version	License	Comment
1	Database	PostgreSQL	18	PostgreSQL License	Reliable open-source relational database for structured application data.
2	Database extension	pgvector	0.8	PostgreSQL License	Adds vector storage and similarity search directly inside PostgreSQL
3	Programming language / runtime	Python	3.11	Python Software Foundation License Version 2	Core runtime for backend
4	Database ORM / SQL toolkit	SQLAlchemy	2	MIT	Provides Python ORM and SQL toolkit support for database access and queries.
5	Backend API framework	FastAPI	0.136	MIT	Build easy python basd APIs
6	Data validation	Pydantic	2.13	MIT	Validates, parses, and serializes Python data models safely.
7	LLM application framework	LangChain	1.2	MIT	Build LLM workflows
8	NLP library	spaCy	3.8	MIT	Provides production-ready NLP features such as tokenization, NER, and pipelines. Contains e.g. BERT Implementations for NLP tasks like classification
9	ASGI server	Uvicorn	0.46	BSD-3-Clause	Runs the FastAPI backend
10	PostgreSQL async driver	asyncpg	0.31	Apache License 2.0	Provides fast asynchronous PostgreSQL connectivity for Python services.
11	LLM observability	Langfuse	4.5	MIT Expat for core; enterprise directories under separate ee/LICENSE	Tracks LLM traces, prompts, latency, costs, and evaluation data.
12	HTTP client	HTTPX	0.28	BSD-3-Clause	Provides sync and async HTTP calls to model APIs and backend services.
13	Testing framework	pytest	9	MIT	Automated Testing
14	Environment configuration	python-dotenv	1.2	BSD-3-Clause	Loads local environment variables from .env files during development.
15	Tokenization	tiktoken	0.12	MIT	Counts and encodes tokens. Used by LangChain for Textsplitting.
16	LLM frontend prototyping	Streamlit	1.56	Apache License 2.0	Quickly builds Python-based web apps, dashboards, and chat prototypes for LLM applications. Used for protoyping & early previews.
17	Large language model	Gemma	Gemma 4 31B	Apache License 2.0	Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.

#	Context	Name	Version	License	Comment
18	Large language model	Qwen	Qwen3.5-35B-A3B	Apache License 2.0	Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.
19	Large language model	Mixtral	Mixtral 8x7B	Apache License 2.0	Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.
20	LLM gateway / provider abstraction	LiteLLM	1.83	MIT for OSS package; enterprise package/features under separate commercial license	Provides a unified OpenAI-compatible interface for calling multiple LLM providers.
21	Containerization / deployment	Docker Engine	29	Apache License 2.0	Packages and runs the application stack consistently across environments.
22	Web framework	Flask	3.1	BSD-3-Clause	Server-side web framework for the production frontend
23	Template engine	Jinja2	3.1	BSD-3-Clause	Template engine used with Flask to generate server-rendered HTML views
24	Chat / users	OpenWebUI	0.9	Open WebUI License	chat-style interaction and user management. Current OpenWebUI versions use the Open WebUI License, which is based on BSD-3-Clause but adds a branding / attribution restriction

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

#	Context	Name	Version	License	Comment

Last Name	First Name	Value					
Kahvecioglu	**Selin**	8		**9.00**	**NOK**		
de Sa Filho	**Fabiano**						
Ammann	**Paul**	13					
Schmidt	**Alexander**						
Wendlandt	**Julian**	8		0	No size		
Müermann	**Ole**	8		1	Trivial size		
Klarer	**Paul**			2	Small size		
				3	Medium size		
Annus	**Mohammad**	8		5	Large size		
Rasul	**Hamid**	8		8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							